curve reproduced by anything that glows because it is hot.[fn11] And it turns out that if you measure the intensity of the cosmic microwave background at different frequencies, you get the most precise, most perfect blackbody curve ever measured in nature. The only way to explain this is that the universe itself was once, everywhere, extremely hot.

Legend has it that the first time this result was presented as a graph in a conference talk, the audience actually cheered. Part of their enthusiasm was certainly because the measurement was extremely impressive and precise, and a perfect fit to the theory (which is always nice to see). But I'm fairly sure that another part was the fact that people realized they were SEEING THE BIG BANG. Actually *seeing* it. I, for one, am still not quite over this.

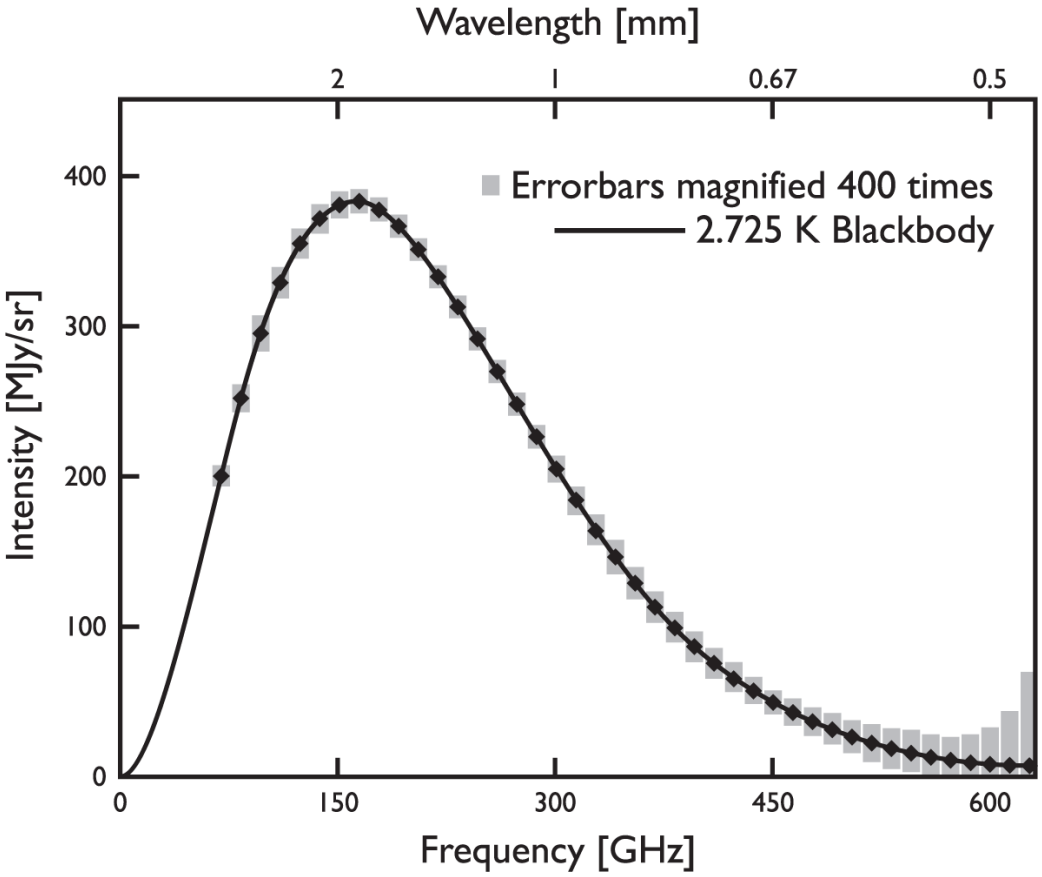


Figure 4: The blackbody spectrum of the cosmic microwave background. The height of the curve indicates the intensity of the radiation at a given frequency or wavelength. The data points are shown with error bars indicating the uncertainties in the measurement, but with the size of the uncertainties magnified 400 times so they're not all completely hidden behind the width of the line, which is the spectrum you'd expect for something glowing at a temperature of 2.725 Kelvins (-270˚C).